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A. Count the Inversion

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: sort, open, file, creat(, fstream, thread, process, system(, exec(

Here is a Pseudocode of the Merge Sort Algorithm.

```
def merge(a, b):
    # write your code here
    # a and b are two sorted List
    # merge function will return a sorted List after merging a and b

def mergeSort(arr):
    if len(arr) <= 1:
        return arr
    else:
        mid = len(arr)//2
        a1 = mergeSort(.....) # write the parameter
        a2 = mergeSort(.....) # write the parameter
        return merge(a1, a2)    # complete the merge function above
```

Now, you are given an array **A** of size **N** of **N** distinct integers. It is guaranteed that the array A contains a permutation of integers from 1 to N (i.e., every integer from 1 to N appears exactly once).

1. Count the number of inversions in the given array.
2. Sort the array in non-decreasing order.

An inversion is a pair (i, j) where $i < j$ and $A[i] > A[j]$.

Input

The first line contains an integer **N** ($1 \leq N \leq 10^5$) — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq N$) separated by spaces.

Output

In the first line, print the total number of inversions in the given array. In the next line, print the array in non-decreasing order.

Examples

input	Copy
5 1 2 5 4 3	
output	Copy
3 1 2 3 4 5	

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→ Last submissions

Submission	Time	Verdict
381801275	Jul/09/2026 08:16	Accepted
381800826	Jul/09/2026 08:10	Wrong answer on test 15
381798731	Jul/09/2026 07:41	Wrong answer on test 3

input	Copy
5 1 2 3 4 5	
output	Copy
0 1 2 3 4 5	

input	Copy
5 5 4 3 2 1	
output	Copy
10 1 2 3 4 5	

input	Copy
7 6 4 2 5 7 3 1	
output	Copy
14 1 2 3 4 5 6 7	

Note

In the first example (1-based indexing), the inversions are the pairs of indices (3, 4), (3, 5) and (4, 5).

In the second example, there are no inversions.

In the third example, every pair of i, j where $i < j$, we have $A[i] > A[j]$. Hence, All 10 such pairs are inversions.

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B. Count the Inversion Revisited

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given an array A of N integers. Find the number of pairs of indexes (i, j) such that $i < j$ and $A[i] > A[j]^2$.

Input

The first line contains an integer N ($1 \leq N \leq 10^5$) — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($-10^6 \leq a_i \leq 10^6$) separated by spaces.

Output

Output a single integer - number of such pairs.

Examples

input	Copy
6 10 2 5 1 -2 25	
output	Copy
6	
input	Copy
5 5 4 3 -2 -1	
output	Copy
4	

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C. Fast Power Drift

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: pow, open, file, creat(, fstream, thread, process, system(, exec(

You are given two integers a and b . Calculate $a^b \bmod 107$.

Input

The input file contains two integers a ($1 \leq a \leq 10^4$) and b ($1 \leq b \leq 10^{12}$).

Output

Print one integer — the result of $a^b \bmod 107$.

Examples

input	Copy
100 3	
output	Copy
85	
input	Copy
100 5	
output	Copy
99	
input	Copy
10000 1000000000000	
output	Copy
27	

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Submission	Time	Verdict
381831042	Jul/09/2026 12:46	Accepted
381830874	Jul/09/2026 12:45	Wrong answer on test 3
381829531	Jul/09/2026 12:33	Accepted

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D. Fast Matrix Drift

time limit per test: 1 second
memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given a 2×2 integer matrix A and an integer exponent X ($1 \leq X \leq 10^9$).

Your task is to compute the matrix A^X , where all intermediate operations (additions and multiplications during matrix multiplication) are performed modulo $10^9 + 7$.

Formally, for two matrices:

$$B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}, \quad C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Their product $D = B \times C$ is defined as: $D_{ij} = (b_{i1} \cdot c_{1j} + b_{i2} \cdot c_{2j}) \bmod (10^9 + 7)$.

Input

The first line contains a single integer T ($1 \leq T \leq 10^5$) — the number of test cases.

The first line of each test case contains four integers a_{11} , a_{12} , a_{21} , and a_{22} where $0 \leq a_{ij} \leq 10^9$ — the elements of the matrix A . The second line contains an integer X where $1 \leq X \leq 10^9$.

Output

For each test case, print two lines, each containing two integers: the resulting 2×2 matrix A^X . Each element should be printed modulo $10^9 + 7$.

Examples

input

Copy

2
1 1 1 0
5
2 0 0 2
3

output

Copy

8 5
5 3
8 0
0 8

input

Copy

3
1 1 1 1
10
1 2 3 4
12
1 2 1 2
5

output

Copy

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Submission	Time	Verdict
382074015	Jul/11/2026 09:11	Accepted
381988310	Jul/10/2026 17:20	Wrong answer on test 2
381988067	Jul/10/2026 17:18	Wrong answer on test 3

```
512 512
512 512
138067399 201223170
301834755 439902154
81 162
81 162
```

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E. Fast Series Drift

time limit per test: 2.5 seconds

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

You are given three integers a , n and m . Calculate $(a^1 + a^2 + \dots + a^n) \% m$.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$) — total numbers of test cases.

In each of the next T test cases, there are three integers a ($1 \leq a \leq 10^6$), n ($1 \leq n \leq 10^{12}$) and m ($1 \leq m \leq 10^9$)

Output

Print one integer — the result of $(a^1 + a^2 + \dots + a^n) \% m$.

Example

input

Copy

```
3
2 5 1000
2 9 1000
1 100 30
```

output

Copy

```
62
22
10
```

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Submission	Time	Verdict
382107165	Jul/11/2026 14:42	Accepted
382105182	Jul/11/2026 14:24	Time limit exceeded on test 2

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F. Ordering Binary Tree

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

you are given an array A of size N in **increasing** order. Find an order of these N integers such that, if these integers are inserted into a Binary Search Tree (BST) one by one, the height of the resulting BST is minimized.

A Binary Search Tree is a binary tree in which each node has at most two children, referred to as the left and right child. For any node, all elements in the left subtree are smaller than the node's value, and all elements in the right subtree are greater than the node's value.

The height of a Binary Search Tree is defined as the maximum depth among all the nodes in the tree.

Note: All the elements in the array A are guaranteed to be unique. In other words, $A_i \neq A_j$ if $i \neq j$.

Input

The first line contains an integer N ($1 \leq N \leq 10^5$) — denoting the length of the list.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq 10^9$) in non-descending order separated by spaces.

Output

Output the order of the elements such that when inserted into a Binary Search Tree, the height of the tree is minimized. If there are multiple such orders then find any of them.

Example

input	Copy
5 1 2 3 4 5	
output	Copy
3 1 2 4 5	

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Submission	Time	Verdict
381844086	Jul/09/2026 14:37	Accepted



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G. 220 Trees

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with N nodes. You are given the in-order and pre-order traversals of the tree. Your task is to determine the post-order traversal of the tree.

Input

The first line contains an integer N ($1 \leq N \leq 1000$) — the number of nodes in the binary tree.

In the next line, there will be N integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq N$) separated by spaces — representing the in-order traversal of the tree.

The following line, there will be N integers $b_1, b_2, b_3 \dots b_n$ ($1 \leq b_i \leq N$) separated by spaces — representing the pre-order traversal of the tree.

Output

Print N space-separated integers representing the post-order traversal of the binary tree.

Example

input	Copy
5 4 2 5 1 3 1 2 4 5 3	
output	Copy
4 5 2 3 1	

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Submission	Time	Verdict
381860561	Jul/09/2026 16:35	Accepted
381857529	Jul/09/2026 16:14	Runtime error on test 1

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H. 220 Trees Reassessed

time limit per test: 1 second

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: open, file, creat(, fstream, thread, process, system(, exec(

There is a Binary Tree with N nodes. You are given the in-order and post-order traversals of the tree. Your task is to determine the pre-order traversal of the tree.

Input

The first line contains an integer N ($1 \leq N \leq 1000$) — the number of nodes in the binary tree.

In the next line, there will be N distinct integers $a_1, a_2, a_3 \dots a_n$ ($1 \leq a_i \leq N$) separated by spaces — representing the in-order traversal of the tree.

The following line, there will be N distinct integers $b_1, b_2, b_3 \dots b_n$ ($1 \leq b_i \leq N$) separated by spaces — representing the post-order traversal of the tree.

Output

Print N space-separated integers representing the pre-order traversal of the binary tree.

Examples

input	Copy
5 1 2 3 4 5 1 4 5 3 2	
output	Copy
2 1 3 5 4	

input	Copy
10 1 2 3 4 5 6 7 8 9 10 2 1 3 5 6 7 4 10 9 8	
output	Copy
8 4 3 1 2 7 6 5 9 10	

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→ Last submissions

Submission	Time	Verdict
381976907	Jul/10/2026 15:49	Accepted
381972725	Jul/10/2026 15:16	Runtime error on test 1
381971775	Jul/10/2026 15:08	Wrong answer on test 1

